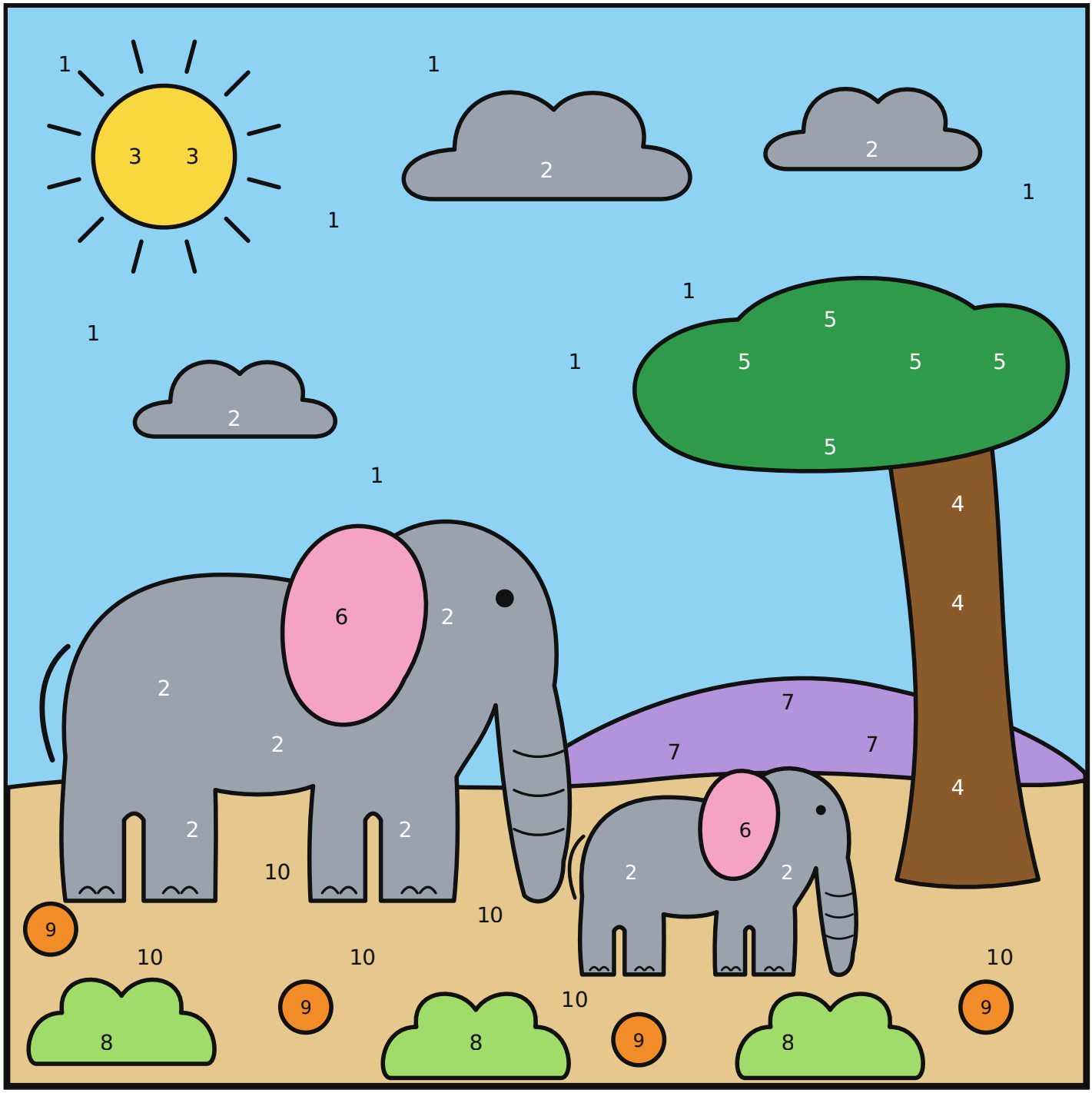
Name: **ANSWER KEY** _____ Date: _____

Similarity: Color By Number



Color Code:

1 Light Blue 9	**4** Brown $\frac{2}{5}$	**3** Yellow $\frac{3}{2}$	**7** Purple 15	**2** Gray 6
10 Tan $\frac{3}{5}$	**6** Pink 21 ft	**5** Green $\frac{4}{25}$	**9** Orange 7.5	**8** Light Green 87.5 mi

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Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.

$\triangle ABC \sim \triangle DEF$. AB = 4, BC = 6 and DE = 6. Find EF.

$\frac{4}{6} = \frac{6}{EF}$

9

1. Color: **Light Blue**

2.

Solve the proportion: $\frac{x}{15} = \frac{4}{10}$

$10x = 60$

6

2. Color: **Gray**

3.

A triangle with a side of 6 is enlarged to a similar triangle with the matching side 9. Find the scale factor.

scale factor = $\frac{new}{old} = \frac{9}{6}$

$\frac{3}{2}$

3. Color: **Yellow**

4.

Two similar polygons have a scale factor of 2 : 5. What is the ratio of their perimeters?

perimeters have the same ratio as the sides

$\frac{2}{5}$

4. Color: **Brown**

5.

Two similar polygons have a scale factor of 2 : 5. What is the ratio of their areas?

areas use the square of the scale factor: $\frac{2^2}{5^2}$

$\frac{4}{25}$

5. Color: **Green**

6.

A 6-ft person casts a 4-ft shadow. A tree casts a 14-ft shadow at the same time. How tall is the tree?

$\frac{6}{4} = \frac{h}{14}$

$4h = 84$

21 ft

6. Color: **Pink**

7.

Rectangle A is 3 by 5. Rectangle B is similar with a short side of 9. Find the long side of Rectangle B.

scale factor = $\frac{9}{3} = 3$

$5 \cdot 3$

15

7. Color: **Purple**

8.

On a map, 1 inch represents 25 miles. Two cities are 3.5 inches apart on the map. How many miles apart are they?

3.5×25

87.5 mi

8. Color: **Light Green**

9.

In $\triangle ABC$, DE ∥ BC with D on AB and E on AC. AD = 4, DB = 6 and AE = 5. Find EC.

$\frac{AD}{DB} = \frac{AE}{EC}$

$\frac{4}{6} = \frac{5}{EC}$

$4 \cdot EC = 30$

7.5

9. Color: **Orange**

10.

Two similar figures have areas 18 and 50. Find the scale factor of the smaller to the larger.

ratio of areas = $\frac{18}{50} = \frac{9}{25}$

scale factor = $\sqrt{\frac{9}{25}}$

$\frac{3}{5}$

10. Color: **Tan**